



* April 5, 2001
✉ mattia.garatti@universite-paris-saclay.fr

Mattia Garatti

Curriculum Vitae

I am a doctoral researcher in Mathematics interested in Mathematical Analysis. Currently, I am carrying out my research on regularized optimal transport problem(s). I am also a private teacher, mainly for high school and university students.

Current position

Doctoral researcher in Mathematics, *Université Paris-Saclay, Orsay*, **2025 – ongoing** France.

Thesis: "On regularized Optimal Transport problems". Supervisors: Luca Nenna - Simona Rota-Nodari.

Education

Master degree in Mathematics (LM-40), *Università Cattolica del Sacro Cuore, Brescia*, **2023 – 2025**

Average: 30 cum laude. Main courses' topics: Real and Functional Analysis, Measure Theory, Differential Geometry, Mathematical Physics. Thesis: "Variational representations of optimal transport and Schrödinger problem". Supervisor: Luca Tamanini - Cosupervisor: Nicolò De Ponti.

Bachelor degree in Mathematics (L-35), *Università Cattolica del Sacro Cuore, Brescia*, **2020 – 2023**

Average: 31.33/30. Thesis: "Il Teorema di esistenza di Peano per le equazioni differenziali ordinarie", provision of a theory on ODEs in the Peano's assumptions. Supervisor: Marco Marzocchi

High School Diploma, *Liceo Scientifico 'Camillo Golgi', Breno (BS)*, **2015 – 2020** 100/100.

Publications

A PDE approach to Benamou–Brenier formula for the Schrödinger problem, **2026**
M. Garatti, L. Nenna, S. Rota Nodari, L. Tamanini, pre-print.

Talks

Entropic regularization(s) of Optimal Transport problem(s), *Università Cattolica del Sacro Cuore, Brescia*, **March, 2026**

Talk about Schrödinger problem with a focus on the Benamou–Brenier formula

Variational representations of optimal transport and Schrödinger problem, *Math PhD Days at UniTO, Turin*, **February, 2025**

Talk on my master thesis work

Research visits

Université Cote d'Azur, Nice. Visited Simona Rota-Nodari	March 16 - March 27, 2026
Università Cattolica del Sacro Cuore, Brescia. Visited Luca Tamanini	March 05 - March 06, 2026
Università Cattolica del Sacro Cuore, Brescia. Visited Luca Tamanini	January 30 - February 06, 2026
Université Paris-Saclay, Paris (France). Visited Luca Nenna	July 28 - August 01, 2025

Teaching

MathExcellence 2025: corso di eccellenza in Matematica , Università Cattolica del Sacro Cuore, Brescia (BS). Assistance and tutoring activities (5 hours)	June 2025
Teacher Assistant, Internship , Liceo Scientifico 'Golgi', Breno (BS). Frontal lectures for high school students, exam administration and correction, laboratory assistant	September 2022
Private Teacher, Internship , Cooperativa 'Assolo', Berzo Inferiore (BS). Lessons of Mathematics and Physics for high school students with SLD	August – September 2018
Teacher Assistant, Internship , Fondazione Scuola Materna 'San Giuseppe', Gratacasolo (BS). Supervision of preschool children, scientific experiments	June 2018

Work experience

Math writer , Zanichelli editore S.p.A., Bologna (BO). Teamwork with the editorial staff, autonomous creation of exercises of various difficulty on different topics to be included in future publications	2024 – 2025
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Summer schools and workshop attended

Modelling Viscoelastic Fluids , Università Cattolica del Sacro Cuore, Brescia. Lecture series about mathematical models for viscoelastic fluids	February 2026
Calculus of Variations and applications , CIRM, Marseille, France. Workshop on the most recent advances in the field	January 2026
Geometry, duality and convexity in new OT problems , IMO Paris–Saclay, Orsay, France. Workshop on new Optimal Transport problems arising in various applications	October 2025
PGMO Days 2025 , EDF Lab Paris–Saclay, Palaiseau, France. Workshop on Optimal Transport and its interactions with Data Sciences	October 2025
Advances in Calculus of Variations and Nonlinear Analysis , Università Cattolica del Sacro Cuore, Brescia. Workshop in honor of Marco Degiovanni's 70th birthday	September 2025
Il Problema di Plateau , Università Cattolica del Sacro Cuore, Brescia. Lecture series about different approach to Plateau problem(s)	April 2025
Introduction to Mechanics of Swelling , Università Cattolica del Sacro Cuore, Brescia. Lecture series about mathematical models for swelling	February 2025
Introduzione ai modelli matematici per i cristalli liquidi , Università Cattolica del Sacro Cuore, Brescia. Lecture series about mathematical models for liquid crystals	February 2024

	Interuniversity Mathematics Summer School, University of Perugia. July – August 2023 Complex Analysis and PDE courses with related final exams successfully passed	
	Mathematics in Ljubljana, University of Ljubljana, FMF and IMFM. July 2023 Summer School with courses on Formalization of Mathematics and Introduction to Complex Dynamics	
Awards	"Smart" study award, Università Cattolica del Sacro Cuore, Brescia (BS). 2025 Award for 100 best students of the university	
	Merit scholarship, Tenaris, Dalmine (BG). 2017 – 2025 Scholarships awarded on merit every year during high school and university	
	Merit scholarship, Consorzio del Bacino Imbrifero Montano dell'Oglio, Bergamo. 2021 – 2025 Scholarships awarded on merit every year during university	
	Exemption from first year university fees, Università Cattolica del Sacro Cuore, Brescia. 2020 Exemption by merit	
	Merit scholarship, Comune di Costa Volpino (BG). 2020 Scholarships awarded on merit for high school diploma	
Organization	Member of the student jury of Galileo Prize. 2024 Jury for award regarding scientific dissemination	
	Member of Joint Teachers – Students Commission, Università Cattolica del Sacro Cuore, Brescia. 2024 – 2025	
	Staff member of Celebration of Mind, Notte dei Ricercatori, Venite a vedere la Terra che gira. Mathematical popularization days, Università Cattolica del Sacro Cuore	
	Staff member of Disfida matematica. 2023 – ongoing Mathematical team competition for high school students, local phase in Brescia	
	Member of Ufficio di Sezione Elettorale, Comune di Costa Volpino (BG). 2021 – 2025 Scrutineer and president of polling station	
	Parish volunteer, Parrocchia di "San Zenone", Gratacasolo. 2015 – ongoing Organization of summer camps and member of the local choir with budget manager functions	
Computer skills	Programming languages: Python, C, C++	Languages Italian: Mother tongue English: Fluent French: Intermediate
	Mark-up languages: LaTeX, HTML	

I hereby authorize the use of my personal data in accordance to the GDPR 679/16 - "European regulation on the protection of personal data".

May 12th, 2026

Mattia Garatti

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